

Abhilesh Kumar

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Education

National Institute of Technology (NIT) Raipur	2022 – 2026
<i>B.Tech in Biotechnology Engineering</i>	<i>CGPA: 7.11</i>
National Inter College, Maudaha, Hamirpur, Uttar Pradesh	2021
<i>Senior Secondary (Class XII), UP Board</i>	<i>74.8%</i>
National Inter College, Maudaha, Hamirpur, Uttar Pradesh	2019
<i>Secondary (Class X), UP Board</i>	<i>78.6%</i>

Technical Skills

- **Programming:** Python, SQL
- **Machine Learning:** Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Cross-Validation
- **Deep Learning:** Neural Networks, LSTM, Basic Model Training
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Seaborn
- **AI / NLP:** Natural Language Processing, Prompt Engineering, Large Language Models (LLM) Basics
- **Tools:** Jupyter Notebook, Git, GitHub, VS Code, Data Visualization

AI / ML Projects

- Movie Recommender System** *Python, Machine Learning*
- Built a content-based movie recommendation system using the TMDb dataset containing 5,000+ movies.
 - Applied TF-IDF vectorization and cosine similarity to analyze movie metadata including genres, keywords, and cast.
 - Designed a recommendation pipeline that suggests top similar movies based on user input.
 - Improved recommendation relevance through text preprocessing and feature engineering.
- CRISPR Off-Target Prediction** [\[link\]](#) *Machine Learning, Python*
- Developed a machine learning model to predict CRISPR-Cas9 genome editing off-target sites.
 - Used GUIDE-seq and CIRCLE-seq datasets to analyze potential unintended DNA edits.
 - Implemented a Random Forest classification model and evaluated performance using ROC-AUC and accuracy metrics.
 - Performed data preprocessing, feature extraction, and model evaluation to improve prediction reliability.
- Customer Churn Prediction** *Python, Machine Learning*
- Built predictive models to identify customers likely to leave a service using a telecom dataset.
 - Implemented Logistic Regression, Decision Tree, and Random Forest algorithms.
 - Performed data cleaning, feature engineering, and model evaluation using confusion matrix and ROC curve.
 - Achieved more than 85% prediction accuracy using a Random Forest classifier.
- AI Chatbot using LLM** *Python, NLP*
- Developed an AI chatbot capable of answering user queries using Large Language Model APIs.
 - Implemented prompt engineering techniques to improve response accuracy and contextual understanding.
 - Integrated natural language processing techniques for question understanding and response generation.
 - Designed a conversational interface capable of handling multiple user queries effectively.

Experience

Korba Hospital

Diagnostic Lab Intern

- Assisted in clinical diagnostic procedures including blood analysis, urinalysis, microbial culture, and ELISA tests.
- Operated laboratory equipment such as centrifuge, microscope, and biochemical analyzers.
- Maintained patient diagnostic records and supported laboratory data management.

Positions of Responsibility

Team Sahaurya, NIT Raipur

Head Coordinator

- Led and coordinated Team Sahaurya at NIT Raipur, overseeing team activities, planning, and execution.
- Coordinated with team members to ensure effective collaboration and timely completion of assigned responsibilities.
- Supported event planning, team management, and communication across different activities.

Achievements

- Developed interdisciplinary projects integrating Biotechnology with Machine Learning applications.
- Gained hands-on experience with data analysis, machine learning models, and AI-based problem solving.
- Built multiple end-to-end ML projects including recommender systems, predictive models, and AI chatbots.

Languages

English, Hindi